

Project 2: Extended Ray Tracing Engine

CS 440 Computer Graphics
Habib University
Spring 2026

Due: 11:59pm on Monday, 4 May

In this project you will extend your ray tracing engine from Homework 5. You will add more features such as acceleration structures, materials with BRDFs, light sources, samplers and tracers. You will also ray trace an original scene to showcase the features of your ray tracing engine.

Code Requirements

Your ray tracing engine must support the following features.

Appearance

Create a new folder for BRDF and implement **diffuse**, **glossy** and **specular** BRDFs. Define new material subclasses from **Material** that use them.

Lighting

Implement at least **two** light sources, e.g. point, spot or directional. Add a new folder called **lights** and populate it with a hierarchy of **Light** classes. Add an array of lights `std::vector<Light*>` as a member in your **World** class and use it to light the scene.

Acceleration

Add a new folder called **acceleration** and implement at least **one** acceleration structure (e.g. uniform grid, BVH, kd-tree) to speed up your ray tracing algorithm. Add an **Acceleration*** member to **World** and use it to compute ray intersections. Include a flag to render with or without acceleration structure, and mention how to use this flag in your **README** file.

Ray Casting

The ray tracer in **raytracer.cpp** is very basic, it shades objects based only on primary rays. Add a new folder called **tracers** and populate it with a hierarchy of **Tracer** classes. You can start the hierarchy by moving the ray tracer in **raytracer.cpp** to a **Basic** class derived from **Tracer**. Add an additional tracer subclass that implements **shadow rays**. Add a **Tracer*** member to **World** and use it for ray tracing.

Sampling

The basic ray tracer suffers from aliasing as only one ray is cast per pixel, resulting in jagged edges of objects. Implement at least **one additional sampler class** (e.g. random or jittered sampling) under the **sampler** hierarchy that reduces aliasing.

Showcase

This project not only implements the engine but you are required to showcase its features and perform an analysis of its performance.

Scene

Create an *original* scene using your ray tracing engine. It should not be copied from other sources, generated by AI, and should be a product of your own imagination. The scene must contain **1 million primitives**. You may use third-party assets, such as models or textures from other sources. The used assets must be publicly available for free. You can use them to build your original scene but it is not allowed for the whole, or major part of the scene to be reused from somewhere. Alternatively, you can model everything yourself, e.g. using Blender, Maya, or any other 3D modeling software.

You must create two renders of your scene: a low quality render at a resolution of 480x360 and a high quality render with resolution 1920x1080 or higher. You may use different aspect ratios if they better fit your scene. The images must be rendered by your engine. Any post-processing must be implemented within your framework (do not edit them later with Photoshop or any other image editing software). The images do not have to be realistic. You can use features that do not follow real world physics if it better suits your artistic concept. The high quality image must render in less than 12 hours on a modern computer.

Website

You are also required to showcase your project on a website, presenting your scene and implementation details. It should include the following:

- **Scene Overview:** A short description of your scene, how you came up with it, and what techniques you used to build it.
- **Implementation Features:** List all the features you have implemented in your ray tracing engine. This includes changes made for Homework 5 and for this project.
- **Acceleration:** Include a table comparing rendering times of your ray tracer with and without an acceleration structure. Supporting renderings must be included.
- **Build Details:** A list of corresponding `World::build` file for every image included on the page.
- **Contribution:** Clearly describe each team member's contributions.
- **References:** Please include all references used for the project, including any external resources or AI tools.

Bonus

You may be awarded extra points for adding any extra features such as motion blur, depth of field etc., or artistically creative scenes where visual quality is beyond expectations. If you choose to add any extra features, make sure to document them clearly on your webpage with supporting images and corresponding build functions.

Team Requirements

The project will be attempted in teams of 3. If you cannot find a team or have fewer members, please let the course instructor know.

Submission Requirements and Instructions

You will turn in your code via Canvas. Please submit a single zip file containing all the required files. The zip file must be named using HU IDs for all members. For example, if my ID is aa04049 and my partner's aa04050, I would name my zip file as "04049_04050.zip". Please **do not** include your name or initials anywhere in your **code submission** (you may include them on your website). Also, do not create a .rar or .tar file – please create a .zip file. Please include a **README** file with instructions on how and which files to run your project, or any other additional details that may be necessary.

Make sure to follow these guidelines carefully. If you do not follow the submission instructions it is possible that you will get a small point deduction.

Sample Scene Ideas

With this project, you have considerable creative freedom in choosing the theme and design of your scene. Below are a few example ideas from different ray tracing competitions to help you get your creative juices flowing.

- [Internet Ray Tracing Competition](#)
- [Ray Tracing Competition by Stanford](#)
- [ETH Zurich Rendering Competition 2025](#)
- [Utah Teapot Rendering Competition](#)

Tips and Notes

There is no project proposal submission. If you wish to discuss your project idea, please schedule a meeting with the instructor.

You are allowed to use a website template from any online source. If you wish to incorporate 3D assets in your project, a short list is given below. Note that you may have to create custom parsers to work with specific file formats for models available on some of these websites.

- 3D scanning repository at [Georgia Tech](#) or [Stanford](#).
- 3D model repository by [Keenan Crane](#) (CMU)
- [Sketchfab](#)
- [TurboSquid](#)